

Data Sources

Treasury Wallet

Primary Treasury Wallet:

0x5E91B40467FB8902C46A7B6CB90482363188D645

Network:

Arbitrum One

Arbiscan:

<https://arbiscan.io/address/0x5e91b40467fb8902c46a7b6cb90482363188d645>

Requirements:

Fetch:

- Current token balances
- Treasury asset allocation
- Recent transfers
- Transaction count
- Wallet age (based on first transaction)
- Counterparty activity
- Treasury value in USD

Preferred providers:

1. Alchemy Portfolio API
2. Covalent API
3. Arbiscan API

Abstract all treasury functionality into:

`services/treasury.ts`

Variational API

Base URL:

<https://api.variational.io>

Documentation:

<https://docs.variational.io/technical-documentation/api>

Primary Endpoint:

GET /metadata/stats

Use this endpoint as the primary protocol data source.

Extract and normalize:

- Total Value Locked (TVL)
- Total Open Interest
- 24h Volume
- Cumulative Volume
- Markets
- Funding Rates
- Mark Prices
- Long Open Interest
- Short Open Interest
- Bid / Ask information
- Loss Refund Pool
- Loss Refunds Paid

Create a typed service:

services/variational.ts

All API responses should be normalized before reaching UI components.

Price Data

Provider:

CoinGecko API

Use for:

- Token USD prices
- Treasury valuation
- Asset allocation calculations

Create:

Required TypeScript Types

Create strongly typed interfaces for:

ProtocolStats

Market

FundingRate

TreasuryAsset

TreasurySummary

TreasuryTransfer

AssetAllocation

PriceData

AnalyticsMetrics

Never use "any".

All services should return typed objects.

Analytics Calculations

Implement client-side calculations:

Treasury Coverage Ratio

$\text{treasuryValue} / \text{totalOpenInterest}$

Treasury to TVL Ratio

$\text{treasuryValue} / \text{tvl}$

Loss Pool Coverage Ratio

$\text{lossRefundPool} / \text{totalOpenInterest}$

Market Concentration Ratio

$\text{largestMarketOI} / \text{totalOpenInterest}$

Long / Short Ratio

$\text{totalLongOI} / \text{totalShortOI}$

Largest Treasury Position

$\text{largestAssetValue} / \text{treasuryValue}$

Asset Allocation Percentage

$\text{assetValue} / \text{treasuryValue}$

Refresh Strategy

Protocol Stats: Refresh every 60 seconds

Treasury Holdings: Refresh every 60 seconds

Treasury Transfers: Refresh every 30 seconds

Price Data: Refresh every 60 seconds

Use TanStack Query caching and automatic background refetching.

Expected Dashboard Feel

The finished product should look closer to:

- Hyperliquid Analytics

- Arkham Intelligence
- Token Terminal
- DefiLlama Pro
- Bloomberg Terminal
- Coinbase Prime

and not like:

- Generic admin templates
- CRM dashboards
- SaaS dashboards

The design should immediately communicate:

"Professional crypto protocol analytics platform."

Use high information density, premium spacing, subtle motion, modern charting, institutional-grade typography, and exceptional visual hierarchy.